

SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – [SIH26090](#)
- **Problem Statement Title** – AI-Driven Voice-to-Commerce, Multilingual Cataloging & Sovereign Provenance Platform for Rural & Tribal Artisans
- **Theme** – Smart Automation / Inclusive Rural Development / Heritage & Culture
- **PS Category** – [Software](#)
- **Team ID** – [\[Insert Your Team ID\]](#)
- **Team Name (Registered)** – [\[Insert Your Team Name\]](#)





KALA KART (कला कार्ट)

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed Explanation

- **10-Sec Voice Listing:** Artisans speak in native vernacular dialects; system auto-extracts craft title, materials, story, and specs.
- **Authentic Voice Preservation:** Genuine recorded artisan voice notes are permanently preserved and streamed via interactive buyer soundwave players.
- **CV AI Studio (5500K):** Transforms poor village hut smartphone photos into studio-grade catalog shots with mineral dye calibration.
- **Fair Wage Pricing Engine:** Computes fair retail prices using Ministry of Labour daily skilled wage benchmarks.
- **India Post Doorstep Pickup:** Automated Speed Post booking directly from rural pin codes.

Addressing the Problem

- **Overcomes Digital Illiteracy:** Zero typing or complex English menus; 100% voice-first interface in regional dialects (Bhojpuri, Maithili, etc.).
- **Eliminates Middlemen:** Replaces exploitative aggregators taking up to 80% margins, returning **85%+ Direct Payout** to artisans.
- **Solves Photo Rejections:** Eliminates commercial photography constraints without expensive camera equipment or studios.
- **Transparent Remittance:** Funds disburse straight into PM Vishwakarma linked DBT accounts.

Innovation & Uniqueness

- **Human-Touch Audio Commerce:** Urban buyers connect emotionally with the real artisan's voice and heritage story.
- **6 Sovereign Pillars:** Interoperable with PM Vishwakarma, GI Registry, Bhashini (MeitY), India Post, SFURTI, and Kala Sakhi.
- **Cryptographic GI Provenance:** SHA-256 tamper-proof QR certificates authenticating regional craftsmanship.
- **Zero Paid SaaS Stacks:** Built on open-source Python, FastAPI, SQLite, and serverless architectures.



TECHNICAL APPROACH

Technologies to be Used

- **Frontend & UX:** Mobile-first responsive UI, HTML5 **MediaRecorder API**, Dual-Language (Hindi & English), Feather Icons.
- **Backend & Microservices:** Python 3.10+, **FastAPI (ASGI)**, Uvicorn, Mangum (AWS Lambda & Vercel Serverless Ready).
- **Database & Cache:** SQLite with ACID compliance, WAL mode, JSON fields, and safe **/tmp** serverless replication.
- **AI & Speech Models:** **Bhashini Open APIs** (AI4Bharat IndicTrans2 & IndicConformer ASR/TTS) for 22 Indic dialects.
- **Computer Vision Studio:** PIL/Pillow (5500K daylight white balancing, micro-contrast enhancement, natural dye swatches).
- **Logistics & KYC:** India Post CEPT API bridge, QR Code generator, PM Vishwakarma e-KYC integration.

Methodology & Process

- **Audio Timeslicing (250ms):** Continuous streaming audio capture with format transcoding (.webm, .ogg, .mp4).
- **Dialect Normalization:** Real-time speech-to-text with bilingual entity extraction (craft type, colors, dimensions).
- **Photometric Image Correction:** Mathematical luminance curves removing harsh village hut shadows without bleaching pigments.
- **Statutory Wage Formula:** $\text{Price} = (\text{Days} \times \text{Wage}) + \text{Materials} + \text{GI_Margin}$ calibrated against Ministry benchmarks.
- **Status:** **100% Working Prototype Live & Deployed**.

End-to-End Implementation Workflow:





FEASIBILITY AND VIABILITY

Feasibility Analysis

- **Technical Feasibility:** Ultra-lightweight footprint (<150ms API latency); runs smoothly on low-bandwidth 2G/3G mobile networks.
- **Economic Feasibility:** Built entirely on open-source stacks with zero licensing or subscription fees, ensuring near-zero per-listing transaction costs.
- **Operational Feasibility:** Leverages existing national grassroots infrastructure—India Post's 155,000+ branch post offices and CSC VLE networks.
- **Deployment Readiness:** Dockerized, serverless-compatible microservices deployable on NIC Cloud / MeghRaj.

Potential Challenges & Risks

- **Dialect & Acoustic Noise:** Heavy rural background noise (cattle, machinery, wind) and non-standard colloquial dialects.
- **Poor Camera Optics:** Motion blur, harsh shadows, and lens distortion from entry-level low-cost smartphones.
- **Digital Hesitation:** Reluctance and fear of technology among elderly, illiterate generational artisans.
- **Internet Intermittency:** Frequent cellular network dropouts in deep tribal belts.

Mitigation Strategies

- **Dialect Noise Filtering:** Client-side audio waveform validation paired with Bhashini IndicConformer noise-filtering pipelines.
- **Photometric Equalization:** Dynamic CLAHE algorithms that auto-bracket exposure and correct chromatic distortions.
- **"Kala Sakhi" Model:** Doorstep assisted onboarding by trained village women SHG leaders (VLEs) who visit homes.
- **Offline-First Caching:** Local SQLite & IndexedDB storage queues orders and voice recordings when offline.



IMPACT AND BENEFITS

Potential Impact on Target Audience

- **Empowers 7+ Million Artisans:** Direct digital inclusion for rural, tribal, SC/ST, and PM Vishwakarma craftspeople across 700+ districts.
- **Income Multiplication (2.5x – 3.5x):** Bypasses exploitative middlemen, recapturing economic value directly for rural households.
- **Financial Dignity & Formal Credit:** Creates transparent digital sales ledgers, enabling artisans to qualify for collateral-free MUDRA loans.
- **Generational Craft Preservation:** Encourages youth in artisan families to continue ancestral trades through viable livelihoods.

Multi-Dimensional Benefits

- **Economic Benefits:**
 - Guaranteed **85%+ Direct Payout** straight into Aadhaar-linked bank accounts.
 - Transparent pricing breakdown eliminates seasonal distress selling.
- **Social & Cultural Benefits:**
 - Safeguards endangered GI crafts (Madhubani, Dokra, Bidriware).
 - Empowers rural women who comprise over 56% of the craft workforce.
- **Environmental Benefits:**
 - Champions 100% biodegradable crafts and natural mineral dyes over petrochemical mass plastics.

85%+

Direct Wage Guarantee

250–350%

Artisan Income Increase

155,000+

India Post Rural Access Nodes

22 Languages

Bhashini Indic AI Support



RESEARCH AND REFERENCES

Government Policies & Statues

- **Ministry of MSME:** *PM Vishwakarma Scheme Operational Guidelines (2023–2028)* — Framework for artisan verification and digital toolkits.
- **Ministry of Commerce & Industry (CGPDTM):** *Geographical Indications of Goods Act, 1999* & Official Intellectual Property India GI Registry.
- **Ministry of Electronics & IT (MeitY):** *National Language Translation Mission (NLTM)* — Bhashini Open API Standards & IndicTrans2 Benchmarks.
- **Ministry of Labour & Employment:** *Office of Chief Labour Commissioner (Central Sphere)* — Skilled Handicraft Minimum Wage Index 2025/26.

Technical Standards & Research Work

- **Department of Posts (India Post):** CEPT Mysore Speed Post API & Merchant Pickup Integration Protocols.
- **Open Network for Digital Commerce (ONDC):** Beckn Protocol Open E-Commerce Specifications for Decentralized Commerce.
- **AI4Bharat Research (IIT Madras):** *IndicConformer* & *IndicTTS* — Speech Recognition and Synthesis across Indian Languages.
- **Live Project Repository:** <https://github.com/bhavik014/kala-kart>

Verification & Demonstration Artifacts:

- Fully functional web application running with live voice recording, 5500K studio rendering, and Bhashini translation.
- Seeded with 35 authentic Indian GI-crafts across Madhubani, Terracotta, Silk, Bidriware, and Woodcraft.
- Interactive SIH Pitch Presentation Deck accessible directly via /presentation route.